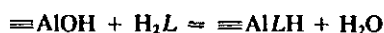


Acid/Base Properties and Phenylphosphonic Acid Complexation at the Aged γ - Al_2O_3 /Water Interface

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Received January 30, 1995; accepted April 18, 1995

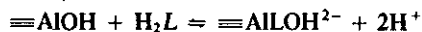
Acid/base properties and phenylphosphonic acid (H_2L) complexation at the water-suspended-and-aged γ - Al_2O_3 /water interface have been studied in 0.1 M NaCl medium at 25.0°C in the range $5.0 < -\log[\text{H}^+] < 9.5$. Equilibrium measurements were performed as a series of potentiometric titrations supplemented with spectrophotometric phosphorous analyses. In the evaluation of the experimental data, the contribution from electrostatic forces was accounted for by using the constant-capacitance model. The adsorption of phenylphosphonic acid onto the studied phase was found to be strong at $-\log[\text{H}^+] < 7$. Desorption was observed when $-\log[\text{H}^+]$ was increased toward 9.5. The specific capacitance was evaluated from data collected in the absence of phenylphosphonic acid. The model describing the acid/base reactions and phenylphosphonic acid binding onto the studied alumina surface, ($\equiv\text{AlOH}$), is given by the following equilibria, intrinsic constants, specific capacitance, and site density:



$$\log \beta_{011(\text{int})} = 9.58 \pm 0.18$$



$$\log \beta_{-111(\text{int})} = 2.94 \pm 0.08$$



$$\log \beta_{-211(\text{int})} = -5.30 \pm 0.13$$

$$C = 1.40 \text{ F m}^{-2}$$

$$\text{Site density} = 1.03 \text{ sites/nm}^2$$

In a series of modeling calculations, the complexation features of phenylphosphonic acid are compared to those of orthophosphoric acid. © 1995 Academic Press, Inc.

Key Words: γ - Al_2O_3 ; acid/base properties; alumina; aluminum

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oxide; constant-capacitance model; adsorption; equilibrium constant; solid-solution interface; phenylphosphonic acid; potentiometric titration; surface complexation.

INTRODUCTION

The surface properties of metal oxides have been the subject of many investigations during recent years. It has, for instance, been acknowledged that metal oxides suspended in natural waters play an important role for the speciation and transport of various naturally occurring substances (1).

The surface chemistry of aluminium oxide, Al_2O_3 , has been of interest, not only because it is one of the most abundant oxide minerals in natural systems, but also because it constitutes the surface layer in a wide variety of industrial materials. For example, the surface chemistry of γ - Al_2O_3 is of special interest to the capacitor industry, since the dielectricum in an electrolytic capacitor consists of a thin film of γ - Al_2O_3 (2). The homogeneity and stability of this aluminium oxide layer are important factors in the service life and electric properties of the capacitor. It has been shown that these qualities can be improved by allowing suitable adsorbates to bind to the surface. The adsorbate most commonly used today is the orthophosphate ion. However, considering that predominantly water containing organic electrolyte solvents are being used, stable organophosphorous compounds might be a better choice. A basic requirement for the success of such a substitution is of course that the ligand chosen binds to the aluminium oxide surface with high affinity.

In the present work acid/base properties and phenylphosphonic acid (H_2L) complexation at the aluminium oxide/water interface have been investigated. The alumina phase used in this study was powdered γ - Al_2O_3 , aged for more than 1 month in ionic medium.

The adsorption reactions recorded in this work are modeled in terms of surface complexation. This implies that the

complexation reactions are described quantitatively by mass law equations. The effect of changes in surface charge, resulting due to the sorption reactions, was thereby accounted for by an electrostatic correction factor. To compare the complexing properties of phenylphosphonic acid with orthophosphoric acid, computerized modeling calculations under different conditions were performed.

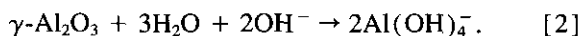
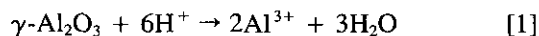
MATERIALS

Chemicals and Solutions

The alumina used in this study was fine powdered high purity, 99.995% γ - Al_2O_3 manufactured by Sumitomo Chemical Co., Ltd. The specific surface area was $140 \text{ m}^2/\text{g}$ according to the BET method. Phenylphosphonic acid (Aldrich) was dried at 100°C overnight and used without further purification. Sodium chloride (Merck p.a.) was dried at 180°C overnight. No further purification was done before use. Dilute solutions of hydrochloric acid were prepared from HCl (Merck p.a.) and standardized against Tris-(hydroxymethyl)aminomethane, "Trisma-base" (Sigma p.a.), which was dried at 105°C overnight before use. Sodium hydroxide solutions were prepared from saturated (19 M) NaOH solution. Crystalline NaOH (EKA Nobel p.a.) was used for the preparation of this concentrate. Dilute NaOH solutions were standardized against standardized acids.

Alumina stock suspensions were prepared to gain $20.00 \text{ g } \gamma\text{-Al}_2\text{O}_3/\text{dm}^3$ suspension. The method used to determine the total concentration of surface hydroxyl groups was a modification of a procedure previously described by Hohl and Stumm (3). The basic idea in this method was to determine the number of protons that could be adsorbed to, or desorbed from, the surface hydroxyl groups when the surface was exposed to low or high $-\log[\text{H}^+]$. ($\equiv\text{AlOH} + \text{H}^+ \rightarrow \equiv\text{AlOH}_2^+$ or $\equiv\text{AlOH} + \text{OH}^- \rightarrow \equiv\text{AlO}^- + \text{H}_2\text{O}$, respectively.) By adding a known amount of strong acid or base to the suspension, analyzing the amount not reacted, and assuming the reactions above to be driven to the far right, the surface concentration could be determined.

To be successful in this determination several corrections have to be considered. When exposed to excess of acid or base, dissolution of alumina takes place according to the reactions



This implies that the dissolved amount of alumina reduces the excess of acid or base remaining in the test suspension.

The dissolution of $\gamma\text{-Al}_2\text{O}_3$ also reduces the particle concentration and therefore causes a reduction of active surface sites. However, since the dissolution reaction rates are much

lower than those of the acid/base reactions at the surface, the two effects can be separated by means of a kinetic procedure. To this end, a series of identical test suspensions were prepared and allowed to equilibrate for different times with excess of acid or base.

The reaction times ranged from 3 to 20 h. By plotting the resulting surface concentration obtained for each test suspension versus the reaction time, it was found that a maximum in surface concentration appeared after a reaction time of 8 h, after which a linear decrease was observed. This is a clear indication that while the proton adsorption/desorption reaction is more or less complete after 8 h, the dissolution of alumina is still continuously decreasing the analyzed surface concentration. The best estimate for the surface concentration was therefore obtained by extrapolating this linear part of the plot, i.e., reaction time of $>8 \text{ h}$, to time zero. The resulting surface concentration obtained from both low and high $-\log[\text{H}^+]$ was $0.24 \text{ mol}\cdot\text{kg}^{-1} \gamma\text{-Al}_2\text{O}_3$. This corresponds to a surface site density of 1.03 nm^{-2} .

The stock solutions of phenylphosphonic acid were prepared by dissolving appropriate amounts of solid to obtain ligand solutions of around 5 mM . Standardizations of these solutions were combined with the determination of acidity constants of the ligand. This means that the acidity constants and the ligand concentration were simultaneously optimized on a series of potentiometric titration data.

All solutions were prepared in a laboratory thermostated to $25.0 \pm 0.5^\circ\text{C}$. Only distilled and boiled water was used. A constant ionic strength of $0.1 \text{ M Na}(\text{Cl})$ background electrolyte was maintained in all solutions by dissolving appropriate amounts of NaCl.

Apparatus

The potentiometric titrations were performed with an automatic system for precise emf titrations. The titrator was a modernized version of the titrator described by Ginstrup (4).

The cell arrangement was similar to that described by Forsling *et al.* (5). The measurements were performed using glass electrodes of the type Ingold NS-201. Ag/AgCl electrodes were prepared according to Brown (6).

The spectrophotometer used in the analysis of phosphate was a Shimadzu UV-2100, UV-visible spectrophotometer.

EXPERIMENTS

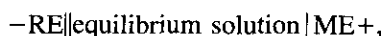
Titration

This investigation has been conducted as a series of potentiometric titrations. In order to minimize variations in activity coefficients, a constant ionic medium of $0.100 \text{ M Na}(\text{Cl})$ was used.

During the titrations, the test solution vessel was im-

mersed in an oil thermostat kept at $25.00 \pm 0.05^\circ\text{C}$. An inert atmosphere was maintained by a continuous flow of argon gas above the suspension in the closed titration vessel. During all experiments suspensions were kept homogeneous by means of a stirring propeller situated a few millimeters from the bottom of the vessel.

The free H^+ concentration, h , was determined by measuring the emf, E , of the cell



in which ME denotes a glass electrode and RE a $\text{Ag}, \text{AgCl}|0.1\text{ M NaCl}$ medium reference electrode. In the titrations, h was varied by addition of H^+ (HCl) or OH^- (NaOH or coulometrically generated OH^-).

By assuming the activity coefficient of H^+ to be constant, h can be calculated according to the formula

$$E(\text{mV}) = E_0(\text{mV}) + 59.157(\text{mV})\log h + E_j(\text{mV}). \quad [3]$$

In this equation, E_0 is a constant determined before and after each titration using a separate solution with known h concentration. E_j is the liquid junction potential, given by the equation

$$E_j(\text{mV}) = -511.5(\text{mV } M^{-1}) \cdot h + 238.7(\text{mV } M^{-1}) \cdot K_w(M^2) \cdot h^{-1}, \quad [4]$$

where $\log K_w = -13.775$ is the ionic product of water in 0.100 M medium (7).

The potentiometric titrations were performed with $-\log h$ ranging from 5.0 to 9.5. This relatively narrow range was chosen to avoid dissolution of $\gamma\text{-Al}_2\text{O}_3$. The dissolution of Al at $-\log h = 5$ and 9.5 was analytically tested and found to be negligible.

Depending on the actual value of $-\log h$ and the presence or absence of phenylphosphonic acid, equilibration times showed a considerable variation. As a criterion for stable readings, a remaining drift in the measured potential of less than 0.1 mV/h was accepted. Typical equilibration times between each addition were 24 h.

During the initial stage of this work, the reversibility of acid/base reactions of $\gamma\text{-Al}_2\text{O}_3$ suspensions was tested by titrating a freshly prepared suspension repeatedly with acid and base in succession. These studies indicated that the system did not behave reversibly, i.e., when the collected data were plotted as the average number of H^+ reacted per $\equiv\text{AlOH}$ (Z) versus $-\log h$ (see Eq. [16]), a series of shifted curves resulted. However, after aging the $\gamma\text{-Al}_2\text{O}_3$ suspension for about 1 month, reasonable reversibility was

obtained. This probably reflects the fact that $\gamma\text{-Al}_2\text{O}_3$ is not thermodynamically stable in aqueous solution but undergoes a slow surface transformation. In an IR/RAMAN investigation of aqueous $\gamma\text{-Al}_2\text{O}_3$ suspensions (2), it was found that there were apparent changes in both the IR and RAMAN spectra when the suspension was aged for 4 to 5 weeks. Moreover, by X-ray diffraction, an aluminium hydroxide phase in the bayerite polymorph was clearly identified. It is therefore important to point out that the pretreatment of the adsorbent must be considered as a variable in surface complexation studies like this and that the surface properties evaluated in this work are not necessarily identical to those of a fresh, metastable $\gamma\text{-Al}_2\text{O}_3$ surface.

Phosphorous Analysis

The potentiometric data were supplemented with analysis of nonbound phenylphosphonic acid, $L_{(\text{aq})}$, in separate series of test suspensions.

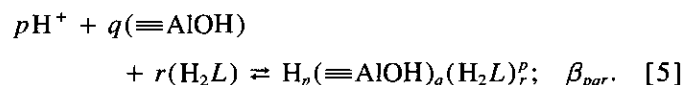
By using the computer program SOLGASWATER (8) together with the preliminary model created using only potentiometric data, suitable test conditions with regard to the $\gamma\text{-Al}_2\text{O}_3$ /phenylphosphonic acid ratio and total H^+ concentration were calculated. These suspensions were prepared in 25 cm^3 Pyrex-glass test tubes and were kept homogeneous under equilibration by means of an end-to-end rotating test tube holder running at approximately 30 rpm.

After 2 weeks of equilibration $-\log h$ in the samples was measured after which the test suspensions were centrifuged and the clear solutions were treated with potassium peroxodisulfate at 120°C for 2 h to oxidize the unadsorbed phenylphosphonic acid to phosphoric acid. Phosphorous was then analyzed according to the molybdenum blue method (9). The intensity of color was measured spectrophotometrically at 880 nm. With this procedure, a total number of 58 different samples has been analyzed.

DATA TREATMENT

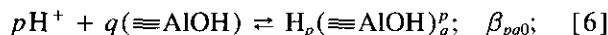
In contact with water, metal oxide surfaces coordinate with water molecules, which leads to a hydroxylated surface (10). In case of alumina, this hydroxylated surface can be expressed as $\equiv\text{AlOH}$. These surface hydroxyl groups can bind and/or release hydrogen ions but also take part in complexation reactions with metal ions and ligands.

Equilibria involving H^+ , $\equiv\text{AlOH}$, and phenylphosphonic acid (H_2L) can be symbolized by the general equation



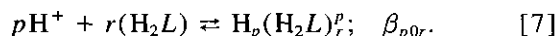
Two special cases of this general reaction are

- (i) the protonation/deprotonation of the alumina surface,



$$\Psi = \frac{T_\sigma \cdot F}{s \cdot a \cdot c} \quad [15]$$

(ii) the deprotonation of phenylphosphonic acid,



It should be noted that β_{pqr} and β_{pq0} are conditional, and must be corrected for by the coulombic energy of the charged surface to obtain the corresponding intrinsic constants

$$\beta_{pqr(\text{int})} = \beta_{pqr} e^{pF\Psi/RT}, \quad [8]$$

where Ψ is the potential at the surface.

The law of mass action and the conditions for the total concentrations of H^+ (H), $\equiv\text{AlOH}$ (B), and H_2L (C) then give the following three equations, in which h , b , and c are the free concentrations of each component respectively.

$$H = h - K_w h^{-1} + \sum p \beta_{pq0(\text{int})} e^{-(pF\Psi/RT)} h^p b^q + \sum p \beta_{p0r} h^p c^r + \sum p \beta_{pqr(\text{int})} e^{-(pF\Psi/RT)} h^p b^q c^r \quad [9]$$

$$B = b + \sum q \beta_{pq0(\text{int})} e^{-(pF\Psi/RT)} h^p b^q + \sum q \beta_{pqr(\text{int})} e^{-(pF\Psi/RT)} h^p b^q c^r \quad [10]$$

$$C = c + \sum r \beta_{p0r} h^p c^r + \sum r \beta_{pqr(\text{int})} e^{-(pF\Psi/RT)} h^p b^q c^r. \quad [11]$$

According to the constant-capacitance model (11), Ψ is related to the surface charge σ by the equation

$$\Psi = \frac{\sigma}{c}, \quad [12]$$

where c is the specific capacitance ($\text{C V}^{-1} \text{m}^{-2}$). The surface charge (in mol dm^{-3}) is obtained from

$$T_\sigma = \sum p \beta_{pq0(\text{int})} e^{-pF\Psi/RT} h^p b^q + \sum p \beta_{pqr(\text{int})} e^{-pF\Psi/RT} h^p b^q c^r \quad [13]$$

or, in electrostatic quantities ($\text{C} \cdot \text{m}^{-2}$),

$$\sigma = \frac{T_\sigma \cdot F}{s \cdot a}. \quad [14]$$

s in this equation is the specific surface area ($\text{m}^2 \cdot \text{g}^{-1}$) and a is the concentration of solid in $\text{g} \cdot \text{dm}^{-3}$. Combining Eqs. [12] and [14] gives

Protonation/deprotonation constants of alumina surface and phenylphosphonic acid were computed using potentiometric titration data. Known parameters are H , B , and C , while h was determined using the emf cell described earlier. With given values of these constants, b and c can be calculated using equations for B and C . These values are then used, together with the measured value of h , to obtain a calculated value of H , H_{calc} . This value is finally compared to the experimental value of H , H_{exp} . By minimizing the error sum of squares, U , defined as $\Sigma(H_{\text{calc}} - H_{\text{exp}})^2$, the equilibrium constants for the protonation/deprotonation reactions, $\beta_{pq0(\text{int})}$ and β_{p0r} , can be obtained as a set of β values resulting in the lowest error sum of squares.

When characterizing the equilibrium model for the three component system, $\text{H}^+ - \equiv\text{AlOH} - \text{H}_2\text{L}$, phosphorous analyses provided, in addition to potentiometric titration data, separate sets of data where also the total concentration of nonbound ligand, $L_{(\text{aq})}$, was known. The calculated value of $L_{(\text{aq})}$, $L_{(\text{aq})\text{calc}}$, can be obtained from the equation for C as a sum of the species $[\text{H}_2\text{L}]$, $[\text{HL}^-]$, and $[\text{L}^{2-}]$ provided that h and the protonation constants of phenylphosphonic acid are known. The computing of the three component system was performed using potentiometric data combined with analyzed nonbound ligand data. Taken over all experimental data, the error sum of squares, U , can now be defined as $\Sigma(W_{\text{emf}}(H_{\text{calc}} - H_{\text{exp}})^2 + W_{\text{analysis}}(L_{(\text{aq})\text{calc}} - L_{(\text{aq})})^2)$. W_{emf} and W_{analysis} are the weighing factors for the emf data and the analyzed nonbound ligand data respectively. To find the "best" equilibrium model, i.e., the model providing the closest fit to experimental data, a systematic variation in the integer values of p , q , and r was performed and, for each choice, the optimization of β_{pqr} constants was performed by a stepwise variation of one or several constants until the values resulting in the lowest error sum of squares was found. The computations were performed with the aid of the computer program LAKE (12).

DATA, CALCULATIONS, AND RESULTS

Acid/Base Properties of Phenylphosphonic Acid

To investigate the deprotonation of phenylphosphonic acid (H_2L), 8 titrations (251 experimental points) were performed. In the measurements $-\log h$ was varied between 2.0 and 8.5 and the total concentration of H_2L ranged from 1.22 to 5.00 mM. The LAKE calculation on these data ended at standard deviation, $\sigma(\text{H}) = 0.01 \times 10^{-3} \text{M}$, giving $\log \beta_{-101} = -1.690 \pm 0.006$ and $\log \beta_{-201} = -8.669 \pm 0.006$.

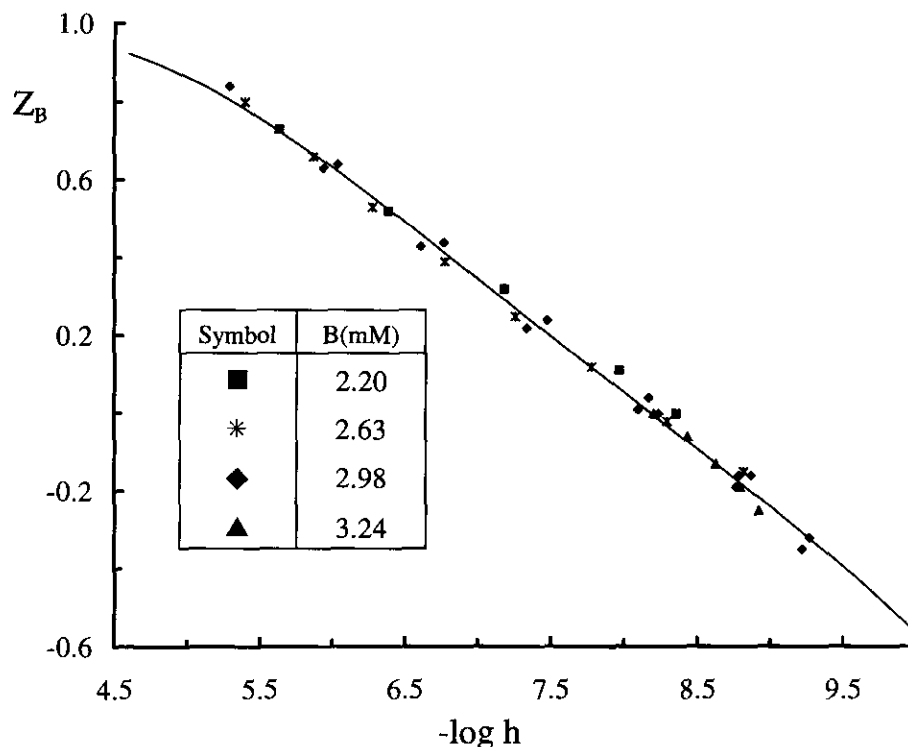


FIG. 1. Experimental data from acid/base titrations in 0.1 M NaCl ionic medium plotted as Z_B versus $-\log h$. Z_B is defined as the average number of protons reacted per $\equiv\text{AlOH}$, $Z_B = (H - h + K_w h^{-1})/B$. The full curve was calculated using the model presented in Table 1.

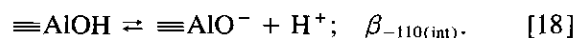
Acid/Base Properties of Water Suspended and Aged $\gamma\text{-Al}_2\text{O}_3$

The acid/base reactions were studied through 4 titrations (33 experimental points) in the range $5.0 < -\log h < 9.5$. To test the reversibility of these reactions, the titrant was changed repeatedly between acid and base in one of the titrations.

The data treatment was based on the experimental points which are illustrated in Fig. 1. In the figure, the average number of H^+ reacted per $\equiv\text{AlOH}$, Z_B , is plotted against $-\log h$. Z_B can be calculated from experimental data as

$$Z_B = (H - h + K_w h^{-1})/B. \quad [16]$$

The model explaining these data includes two intrinsic acid/base constants for the surface hydroxyl groups and an optimized value for the specific capacitance, c . The formation constants are defined according to the following equilibria:



The values obtained for $\log \beta_{110(\text{int})}$, $\log \beta_{-110(\text{int})}$, and c are given in Table 1.

Adsorption of Phenylphosphonic Acid onto Water-Suspended-and-Aged $\gamma\text{-Al}_2\text{O}_3$

Complexation of phenylphosphonic acid was studied in 5 titrations (67 experimental points) and 3 sets of test suspensions (16 experimental points). $-\log h$ was varied between 5.0 and 9.5. The reversibility of these surface reactions were

TABLE 1
Proposed Model in the Three-Component System
 $\text{H}^+ - \equiv\text{AlOH} - \text{Phenylphosphonic Acid (H}_2\text{L)}$

Species	$\log \beta_{pqr} \pm 3\sigma$
HL^-	$\log \beta_{-101} = -1.690 \pm 0.006$
L^{2-}	$\log \beta_{-201} = -8.669 \pm 0.006$
$\equiv\text{AlOH}_2^+$	$\log \beta_{110(\text{int})} = 7.51 \pm 0.09$
$\equiv\text{AlO}^-$	$\log \beta_{-110(\text{int})} = -8.87 \pm 0.11$
$\equiv\text{AlLH}$	$\log \beta_{011(\text{int})} = 9.58 \pm 0.18$
$\equiv\text{AlL}^-$	$\log \beta_{-111(\text{int})} = 2.94 \pm 0.08$
$\equiv\text{AlLOH}^{2-}$	$\log \beta_{-211(\text{int})} = -5.30 \pm 0.13$
Specific surface area	$140 \text{ m}^2\text{g}^{-1}$
Site density	1.03 sites/nm^2
Specific capacitance	$1.40 \text{ F}\cdot\text{m}^{-2}$

Note. The formation constants are related according to the reaction $p\text{H}^+ + q(\equiv\text{AlOH}) + r(\text{H}_2\text{L}) \rightleftharpoons \text{H}_p(\equiv\text{AlOH})_q(\text{H}_2\text{L})_r$. The surface complexes are evaluated on the basis of the constant-capacitance model and the intrinsic constants are given by the equation $\beta_{pqr(\text{int})} = \beta_{pqr} e^{pF\psi/RT}$ (Eq. [8]).

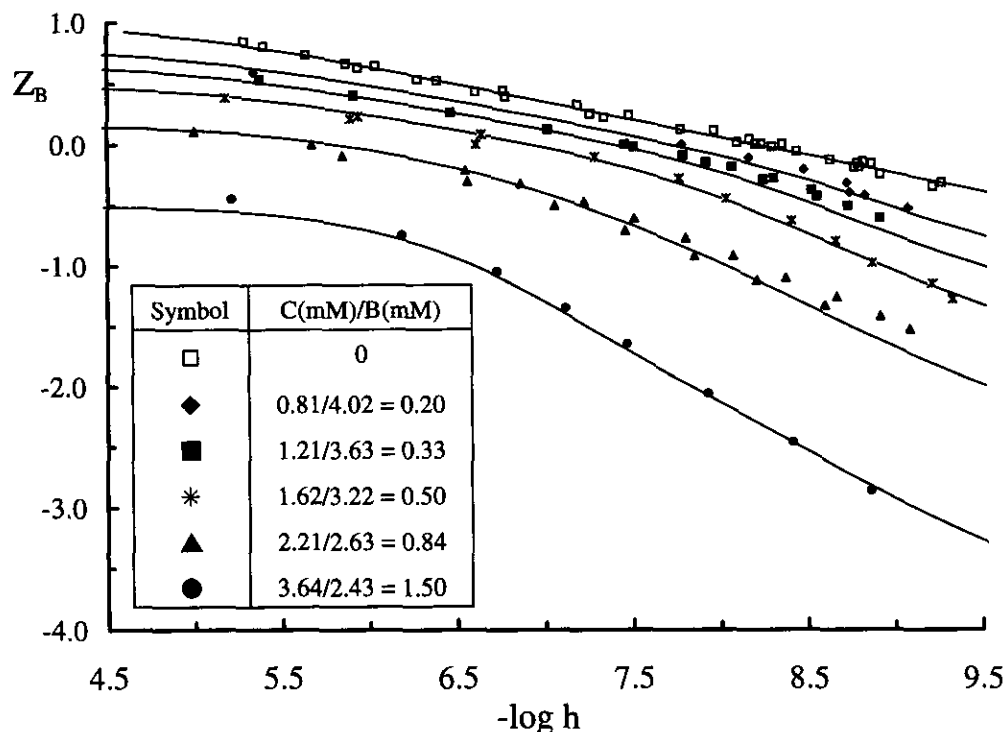
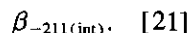
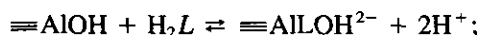
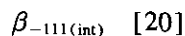


FIG. 2. Experimental data from titrations with γ - Al_2O_3 and phenylphosphonic acid in 0.1 M Na(Cl) ionic medium plotted as Z_B versus $-\log h$. $Z_B = (H - h + K_w h^{-1})/B$. The full curves were calculated using the model presented in Table 1.

tested in titrations in which the titrant was changed repeatedly between acid and base. In the evaluation of equilibrium model no attempts were made to further adjust the acid/base equilibrium constants or the specific capacitance for the alumina surface. The ratio between the total concentration of phenylphosphonic acid (H_2L) and $\equiv\text{AlOH}$ was varied between 0.20 and 1.50. Experimental titration data plotted as Z_B versus $-\log h$ are presented in Fig. 2. The evaluation of the equilibrium model consisted of a test of combinations of complexes with different (p , q , r)-compositions. Both monodentate ($q = 1$) and bidentate ($q = 2$) surface complexes (cf. Eq. [5]) were tested. The best fit to data was given by a model that assumes the formation of three monodentate complexes according to the following reactions:



The corresponding equilibrium constants are given in Table 1.

To illustrate the speciation of phenylphosphonic acid, dis-

tribution diagrams at three different $[\text{H}_2\text{L}]_{\text{tot}}/[\equiv\text{AlOH}]_{\text{tot}}$ ratios were constructed; cf. Fig. 3. These diagrams were constructed using the computer program SOLGASWATER. Contained in these diagrams are also the analyzed data for unadsorbed ligand. As can be seen, the analyzed data correspond well to the model lines for $L_{(\text{aq})}$.

The distribution diagrams indicate also that at $-\log h = 5.5$ and with low $[\text{H}_2\text{L}]_{\text{tot}}/[\equiv\text{AlOH}]_{\text{tot}}$ ratios, one would expect phenylphosphonic acid to be almost stoichiometrically adsorbed to the surface. From the model it also follows that if the ratio between the ligand and the surface sites is increased above unity, all excess of the ligand will remain unadsorbed. This expected behavior was tested in a separate series of batch samples, in which γ - Al_2O_3 suspensions were equilibrated with various amounts of phenylphosphonic acid at $-\log h = 5.5$. The result of this experiment is given in Fig. 4 and shows full agreement with the prediction given by the potentiometric model.

DISCUSSION

The experimental data on which the present investigation is based was collected with $-\log h$ ranging from 5.0 to 9.5. The limits of this rather narrow range was estimated from the aqueous solubility of $\text{Al}(\text{OH})_3(\text{s})$ and set by the necessity to avoid dissolution of aluminium oxide. The amount of dis-

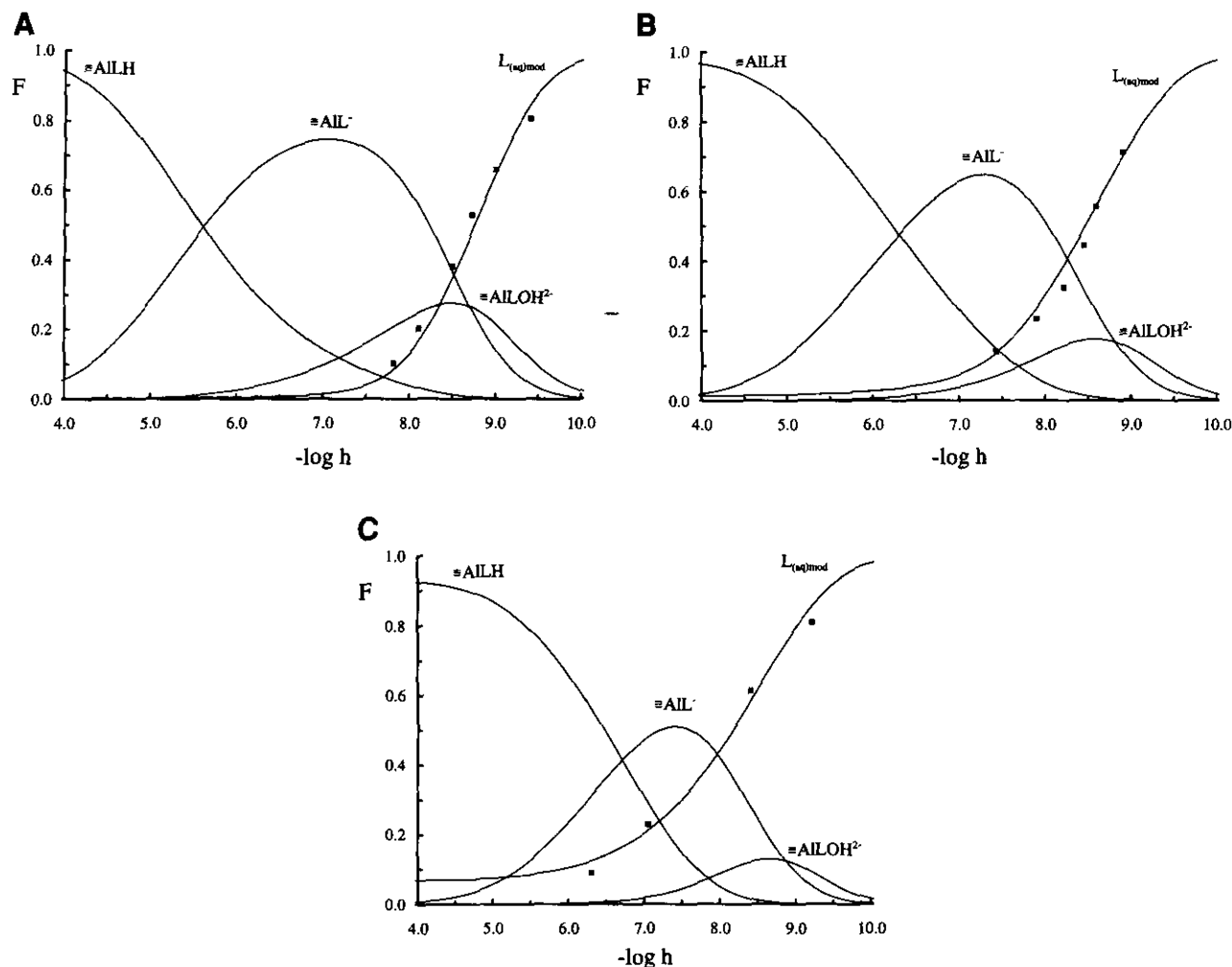


FIG. 3. Distribution diagrams (F versus $-\log h$) at three different $[H_2L]_{\text{tot}}/[\equiv \text{AlOH}]_{\text{tot}}$ ratios, 0.26, 0.53, and 0.78 (A, B, and C, respectively). F is defined as the distribution of phenylphosphonic acid. $L_{(\text{aq})\text{mod}}$ denotes the modeled sum of $[H_2L]$, $[HL^-]$, and $[L^{2-}]$, and the symbol (■) represents experimental data of unadsorbed phenylphosphonic acid.

solved aluminium was also analytically found to be negligible within this $-\log h$ range.

The experimentally determined surface concentration $0.24 \text{ mol} \cdot \text{kg}^{-1}$ corresponds well with the values presented by Hohl and Stumm, $0.25 \text{ mol} \cdot \text{kg}^{-1}$ (3), and Kummert and Stumm, $0.21 \text{ mol} \cdot \text{kg}^{-1}$ (15). Recalculated to an average hydroxyl site density, these data result in site densities close to 1 site/nm^2 . This value is lower than the maximum site density reported in literature, which is as high as 8 sites/nm^2 (13).

The acid/base properties of hydrous $\gamma\text{-Al}_2\text{O}_3$ surfaces have previously been studied by Stumm and co-workers (14, 3, and 15). Their pretreatment procedure included a washing with 0.1 M NaOH and a subsequent storage in distilled water, but no minimum length of storage time before use was stated. Their reported results ($\log \beta_{110(\text{int})} = 7.9; 7.2; 7.4$ and $\log \beta_{-110(\text{int})} = -9.1; -9.5; -10.0$, respectively) show a consid-

erable variation. As described in the section "Experiments," we found $\gamma\text{-Al}_2\text{O}_3$ to be an unstable phase in aqueous suspension that undergoes surface transformation. It is therefore possible that the reported variation in β values is caused by a difference in storage time between the three investigations.

For $-\log h < 7$, the phenylphosphonate ligand was found to adsorb strongly to the alumina surface with the formation of the surface complexes $\equiv \text{AlLH}$ and $\equiv \text{AlL}^-$. With $[H_2L]_{\text{tot}}/[\equiv \text{AlOH}]_{\text{tot}} > 1$, nearly all of the surface sites were occupied by the ligand. At $-\log h \approx 8.5$, minor amounts of a complex $\equiv \text{AlLOH}^{2-}$ was indicated while, with an increase toward 9.5, nearly complete desorption of phenylphosphonic acid was observed. The observation that adsorption is favored in acidic suspensions is common for all anions and weak acids. This situation is opposite to metal ion adsorption, which is favored by alkaline conditions (16).

There are numerous investigations dealing with the ad-

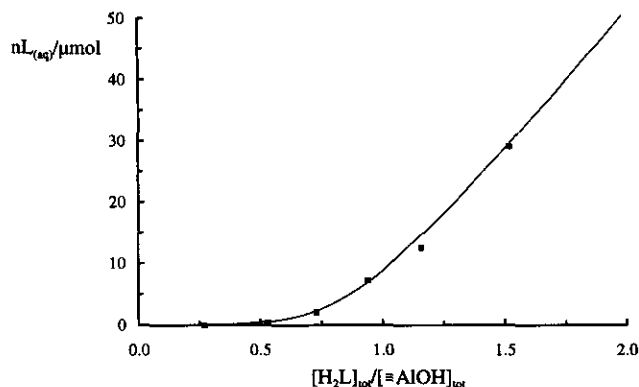


FIG. 4. The amount of unadsorbed phenylphosphonic acid, $L_{(aq)}$ (μmol), as a function of $[\text{H}_2\text{L}]_{\text{tot}}/[\equiv\text{AlOH}]_{\text{tot}}$ at $-\log h = 5.5$. The symbol (■) represents experimental data and the curve was calculated using the equilibrium model.

sorption of orthophosphate ions onto metal oxide surfaces. As a possible explanation of the strong affinity regularly found, it has been suggested that phosphate is bound at two sites on the surface, thereby forming a bridging surface complex. In the present study, no evidence for the formation of bidentate surface complexes was found. The results indicate that the maximum amount of adsorbable phenylphosphonic acid equals the maximum amount of adsorbable protons, and that any excess of ligand remains nonadsorbed (cf. Fig. 4). Our results, however, do not exclude the existence of such bridging surface complexes, provided that one of the coordinating surface sites is "silent" with respect to acid/base properties.

In this study, a surface complex corresponding to the stoichiometry $(-2, 1, 1)$ (pqr) was indicated mainly to better model the analyzed unadsorbed ligand data near the desorption edge. The composition of this minor species, referred to as $\equiv\text{AlOH}^{2-}$ in Table 1, is open to further discussion, since an attachment of an OH^- group to the surface complex $\equiv\text{AlL}^-$, seems unlikely. A possible explanation to the suggested stoichiometry could be that phenylphosphonic acid complexation activates an otherwise "silent" surface hydroxyl group adjacent to the complexation site. This hydroxyl group can thereafter be deprotonated in the high $-\log h$ range.

Ferguson and King (17) have made calculations relating to the solubility of aluminium phosphates in presence of aluminium hydroxide. As a result of these calculations, the possibility was introduced that a phase transformation (i.e., $\text{Al}(\text{OH})_3(\text{s}) + \text{H}_2\text{PO}_4^- \rightleftharpoons \text{AlPO}_4(\text{s}) + \text{OH}^- + 2 \text{H}_2\text{O}$) rather than surface complexation was the phosphorous removal process. Also in the present system, an insoluble Al-phenylphosphonate phase can be suspected to be formed, and we have therefore considered this possibility. Our typical experimental procedure involved the mixing of approxi-

mately 0.20 g Al_2O_3 (i.e., ca. 0.012 mol of $\text{Al}(\text{III})$) with less than 0.00014 mol of phenylphosphonic acid. Therefore, if a phase transformation occurs, the limiting factor will be that of the ligand, and low free ligand concentration will be registered independently of the $[\text{H}_2\text{L}]_{\text{tot}}/[\equiv\text{AlOH}]_{\text{tot}}$ ratio. This is, however, in striking conflict with data presented in Fig. 4, and we can therefore conclude that phase transformations are of no significance in our system at $-\log h = 5.5$. Another very characteristic feature of $-\log h$ dependent phase transformations is that they show a local maximum in buffer intensity at the $-\log h$ of the transformation. An inspection of data presented in Fig. 2 reveals no such behavior and therefore indicates that neither at any other $-\log h$, is the formation of solid Al-phenylphosphonate likely to occur.

Due to the surface complex formation, the surface potential at the alumina surface will vary with $-\log h$ and phenylphosphonate concentration. This is illustrated in Fig. 5 in which the calculated surface potential, Ψ , has been plotted as a function of $-\log h$ and total phenylphosphonic acid concentration. From this figure it can be seen that, when adsorption takes place, the isoelectric point is shifted to lower $-\log h$. When an excess of phenylphosphonic acid is being applied, $[\text{H}_2\text{L}]_{\text{tot}} \geq 0.01 \text{ M}$, the surface becomes negatively charged over almost the entire $-\log h$ range. During the titrations, and when the $-\log h/[\text{H}_2\text{L}]_{\text{tot}}$ conditions were close to the zero surface potential line, as given in Fig. 5, difficulties were encountered in keeping the suspension homogeneous. This can be best explained by a lack of repulsive forces between the particles under those conditions.

To compare the adsorption/desorption behavior of phenylphosphonate with that of orthophosphate, a diagram illustrating the fraction of adsorbed ligand as a function of $-\log h$ was constructed, cf. Fig. 6. This diagram shows that for $-\log h < 8$, binding of phenylphosphonate to the aluminium oxide surface is slightly stronger than the binding of orthophosphate. The figure also shows that the desorption edge

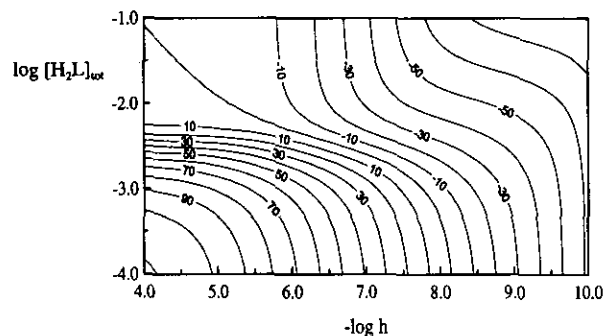


FIG. 5. Surface potential, Ψ (mV), of water-suspended-and-aged γ - Al_2O_3 as a function of $-\log h$ and total concentration of phenylphosphonic acid, calculated at $20.00 \text{ g}\cdot\text{dm}^{-3}$ γ - Al_2O_3 .

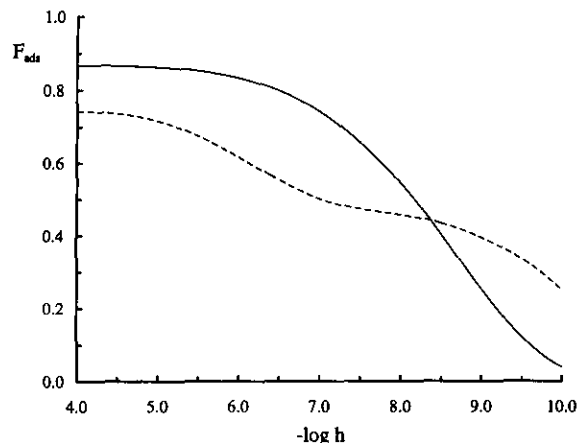


FIG. 6. Fraction of adsorbed ligand in the systems $H^+ - \equiv AlOH$ -phenylphosphonic acid (solid line) and $H^+ - \equiv AlOH$ -orthophosphoric acid (dashed line). The model for the orthophosphoric acid system was recalculated from Huang (18). The calculations are performed at $20.00 \text{ g-dm}^{-3} \gamma\text{-Al}_2\text{O}_3$ and $[P]_{\text{tot}}/[\equiv AlOH]_{\text{tot}} = 1$.

for phenylphosphonate is much sharper than the edge for orthophosphate. This implies that surface complexes of orthophosphate exist to higher $-\log h$ than those of phenylphosphonate, a phenomenon caused by the formation of a species $\equiv AlOPO_3^{2-}$ in the former system. With regard to the possible use of phenylphosphonic acid to stabilize electrolytic capacitors, this calculation shows phenylphosphonate ions to have the most favorable adsorption qualities. This does not necessarily prove them to be superior to orthophos-

phate ions as stabilizers, but it rises the interest to further investigate the electric properties of electrolytic capacitors stabilized with organophosphorous compounds.

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